

CSA0313 - DATA STRUCTURES

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INTRODUCTION

Towers of Hanoi: Three Disk Solution

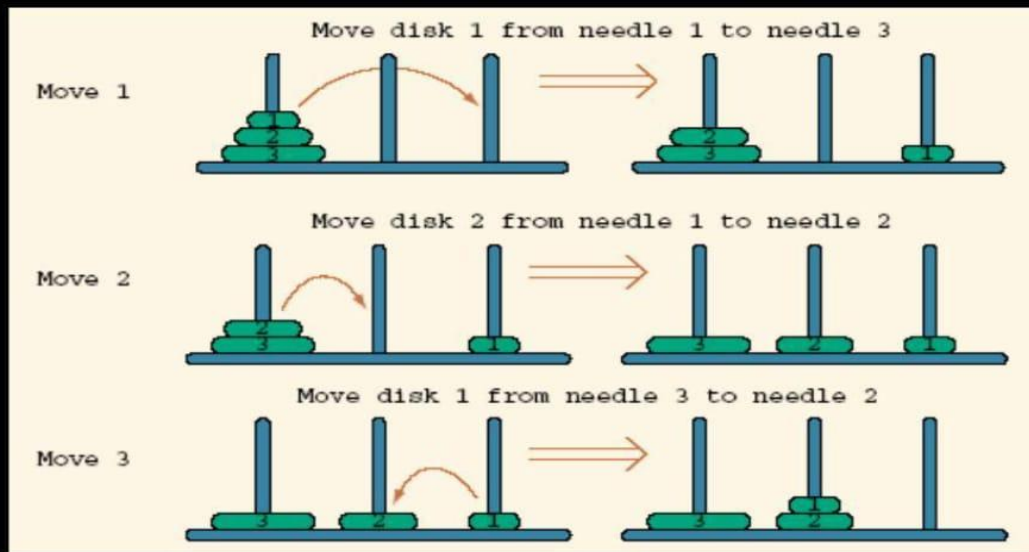
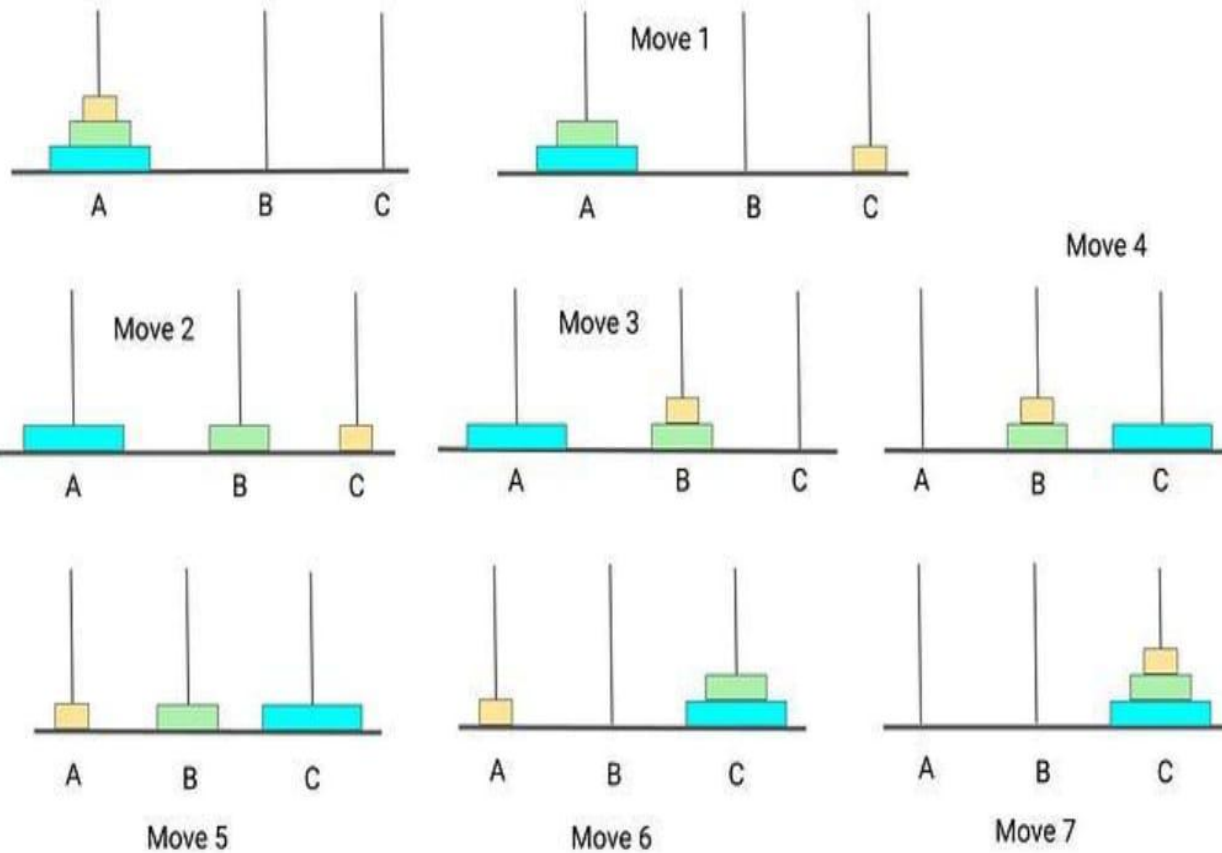


Figure 8: Tower of Hanoi Problem with three disks – Solution 1

- Tower of Hanoi is a famous mathematical puzzle.
- It consists of 3 rods (A, B, C) and n disks.
- The disks are of different sizes and are stacked from largest (bottom) to smallest (top).
- The goal is to move all disks from the Source rod to the Destination rod.

RULES



- **Move only one disk at a time.**
- **Only the top disk can be moved.**
- **Never place a larger disk on a smaller disk.**

ALGORITHM

Tower(n, Source, Auxiliary, Destination)

If $n == 1$

Move disk from Source to Destination

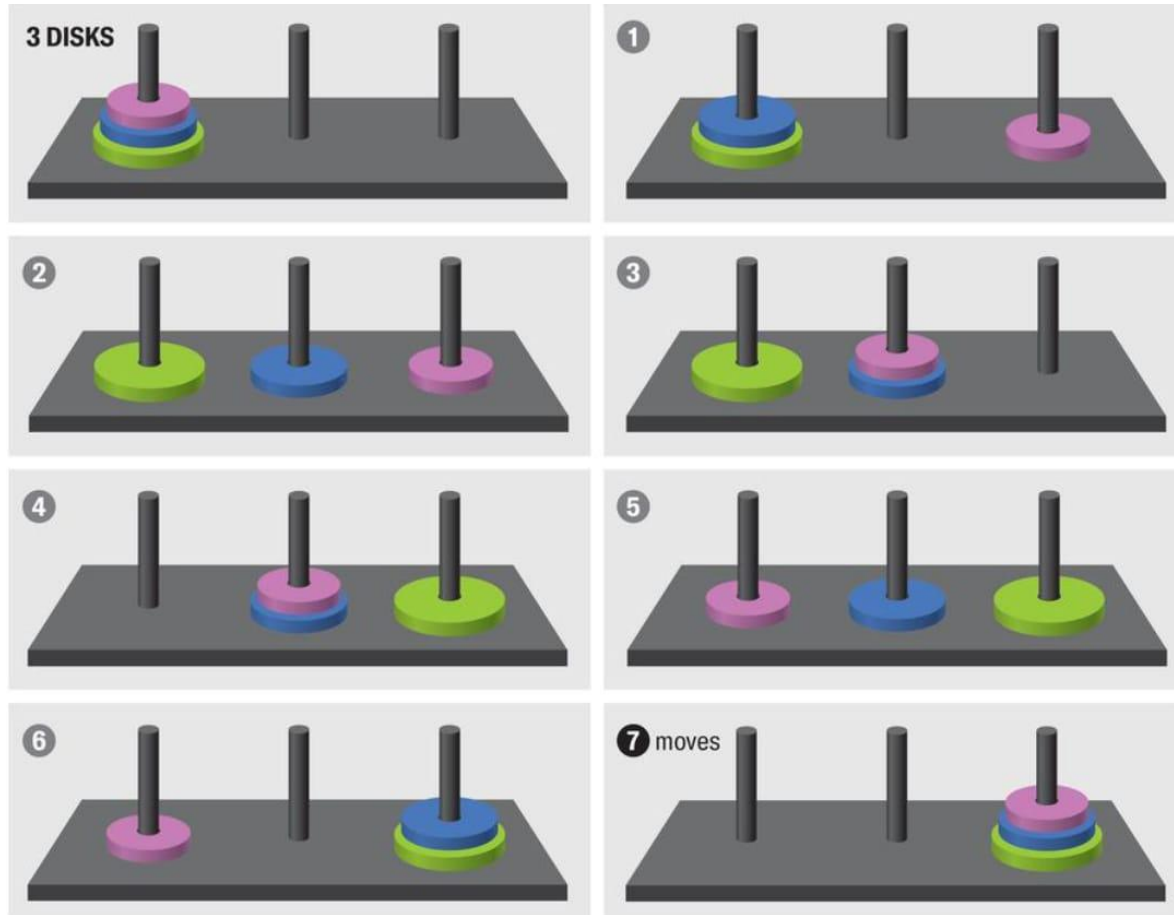
Else

Tower(n-1, Source, Destination, Auxiliary)

Move disk n from Source to Destination

Tower(n-1, Auxiliary, Source, Destination)

FLOW OF EXAMPLE



Move 1: $A \rightarrow C$

Move 2: $A \rightarrow B$

Move 3: $C \rightarrow B$

Move 4: $A \rightarrow C$

Move 5: $B \rightarrow A$

Move 6: $B \rightarrow C$

Move 7: $A \rightarrow C$

Total Moves = 7

PYTHON PROGRAM

```
def tower(n, source, auxiliary, destination):  
    if n == 1:  
        print("Move disk 1 from", source, "to", destination)  
        return  
  
    tower(n-1, source, destination, auxiliary)  
    print("Move disk", n, "from", source, "to", destination)  
    tower(n-1, auxiliary, source, destination)  
  
tower(3, 'A', 'B', 'C')
```

ADVANTAGES

- Easy to understand recursion.
- Improves logical thinking.
- Used to learn recursive algorithms.
- Popular interview and academic problem.

APPLICATIONS

- Teaching recursion.
- Data backup and file transfer concepts
- Artificial Intelligence problem solving
- Algorithm design

CONCLUSION

- Tower of Hanoi is a classic recursive puzzle.
- It demonstrates recursion in a simple way.
- Minimum number of moves required is:
- Minimum Moves = $2^n - 1$